

A Test of Mean Resolution Time

AP Statistics · Unit 4 · Topic 4.5 · B: 2027-01-21 · E: 2027-03-08 · TEACHER KEY

CED TETHER

- **Skill 3.E** — Calculate a test statistic and find a p-value, provided conditions for inference are met.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **Skill 4.E** — Justify a claim using a decision based on significance tests.
- **EU VAR-7** — The t-distribution may be used to model variation.
- **LO VAR-7.E** — Calculate an appropriate test statistic for a population mean, including the mean difference between values in matched pairs. [Skill 3.E]
- **EK VAR-7.E.1** — For a single quantitative variable when random sampling with replacement from a population that can be modeled with a normal distribution with mean μ and standard deviation σ , the sampling distribution of $t = (\bar{x} - \mu) / (s/\sqrt{n})$ has a t-distribution with $n - 1$ degrees of freedom.
- **EU DAT-3** — Significance testing allows us to make decisions about hypotheses within a particular context.
- **LO DAT-3.E** — Interpret the p-value of a significance test for a population mean, including the mean difference between values in matched pairs. [Skill 4.B]
- **EK DAT-3.E.1** — An interpretation of the p-value of a significance test for a population mean should recognize that the p-value is computed by assuming that the null hypothesis is true, i.e., by assuming that the true population mean is equal to the particular value stated in the null hypothesis.
- **LO DAT-3.F** — Justify a claim about the population based on the results of a significance test for a population mean. [Skill 4.E]
- **EK DAT-3.F.1** — A formal decision explicitly compares the p-value to the significance α . If the p-value $\leq \alpha$, then reject the null hypothesis, $H_0: \mu = \mu_0$. If the p-value $> \alpha$, then fail to reject the null hypothesis.
- **EK DAT-3.F.2** — The results of a significance test for a population mean can serve as the statistical reasoning to support the answer to a research question about the population that was sampled.

Essential question: What conclusion is warranted when a one-sample t-test does not cross its evidence threshold?

DOK-3 skill: 4.E · **FRQ pattern:** mean-test-decision-and-conclusion

DOK rationale: Part (c) coordinates the lower-tail p-value, preset threshold, asymmetric decision logic, and SRS population scope to adjudicate two conclusions and trace the consequence of overclaiming.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK: 1**
→ 3 **Minutes: 45**

Teacher does

- Board slide up at the bell. Ask what conclusion the reported evidence permits.
- Begin the 7.5 video follow-along at minute 5.
- Return at minute 33. Require the p-value meaning, decision, scope, and limit.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Commit to one account and cite the numerical comparison or population claim.
- Complete the follow-along, finish (a)–(c), revise, and turn in.

Questions to ask

- What assumption begins a correct interpretation of 0.0645?
- What does 0.0645 greater than 0.05 let us decide, and what does it not establish?
- Which group of tickets can this SRS represent?

Adult role

Solo teacher. Connect the lower-tail probability to the null model and bound the conclusion.

[WARN] WATCH FOR

- Interpreting the p-value as the probability that the null hypothesis is true.
- Using a two-sided p-value even though the alternative is lower-tailed.
- Accepting the null or claiming that the true mean equals 12 hours after failing to reject.

SCORING PART (c) — E / P / I

Expected elements

- **chooses-warranted-account** (required) — Selects the bounded fail-to-reject account
- **compares-p-and-alpha** (required) — Uses $0.0645 > 0.05$
- **interprets-p-conditionally** (required) — Interprets the lower tail assuming $\mu = 12$
- **rejects-proof-of-equality** (required) — Distinguishes failure to reject from acceptance or proof
- **uses-scope-and-consequence** (required) — States the 1,200-ticket scope and both false beliefs

E	Uses the threshold, conditional p-value meaning, scoped conclusion, and both limits of the rejected account.
P	Gives the correct decision but incompletely explains p-value meaning, scope, or why equality was not established.
I	Accepts the null, treats the p-value as a null probability, or concludes the mean equals 12.

[STAR] TODAY'S PROBLEM — annotated

Suppose a help desk closed 1,200 tickets in a quarter. To investigate whether their population mean resolution time was below 12 hours, an analyst took an SRS of 24 and used $\alpha = 0.05$.

Leah: “The lower-tail test gives $t = -1.5747$, $df = 23$, and $p \approx 0.0645$. I would state what evidence this does and does not provide.”

Damon: “Because $p > 0.05$, accept H_0 . The test proves the population mean was exactly 12 hours and the observed difference was only chance.”

Ticket sample

Summary	Value
Population	1200
n	24
Mean (h)	11.1
SD (h)	2.8
Shape	Symmetric; no outliers

FIRST TAKE (5 min)

Which account is more defensible? Cite the p-value comparison or a claim that a test can or cannot justify.

Key: Not graded. Students should distinguish the formal decision from a claim about the null value.

Rules callout “COMPARE EVIDENCE TO THE PRESET THRESHOLD (keep on the page)” is printed on the student sheet.

DOK 1 (a) Define the population mean and state hypotheses for the research question.

Key: Let μ be the mean resolution time for all 1,200 tickets closed that quarter. $H_0 : \mu = 12$ hours and $H_a : \mu < 12$ hours.

DOK 2 (b) Verify random, 10 percent, and shape conditions. Compute SE , t , df , and the lower-tail p-value; interpret the p-value.

Key: The SRS satisfies $24 \leq 0.10(1,200) = 120$; the symmetric, outlier-free sample supports shape for $n = 24$. $SE = 2.8/\sqrt{24} = 0.57155$, so $t = (11.1 - 12)/0.57155 = -1.57467$, $df = 23$, and $p = 0.06449$. If $\mu = 12$, the chance of an SRS result at least this low in the direction of H_a is about 0.0645.

DOK 3 (c) Adjudicate the accounts. Compare p with α , distinguish the formal decision from acceptance or proof, state the scoped population conclusion, and explain the two beliefs produced by the rejected account.

Key: Leah is warranted. Since $0.0645 > 0.05$, fail to reject H_0 . The sample does not provide convincing evidence that the mean for all 1,200 tickets was below 12 hours. The p-value assumes $\mu = 12$; it is not the probability that assumption is true. Failing to reject says the data are not sufficiently incompatible with 12, not that $\mu = 12$ was proved. Chance is plausible but not proved to be the only explanation. The rejected account would falsely report both equality and certainty about chance.

EXIT

One thing the video changed about my first take: _____